National University of Lesotho
Department of Physics and Electronics
EE3304 - Electrical Circuit Analysis
Semester #1

Lab 02: Second-Order Circuits

1 Lab Brief and Purpose

Many phenomena that occur in electric circuits involve or produce time-dependent variables. When a RC, RL, or RLC circuit is suddenly energized or de-energized, a transient phenomenon, which dies out as the circuit approaches it stead-state operation, occurs. This is because of the way in which inductors and capacitors store energy and resistors dissipate it. The exact nature of the transients depends on the values of R, L and C as well as on how they are combined in a circuit. As you progress in your study of circuits you will meet instances where particular transient responses are beneficial and desired and, likewise, instances where such responses are detrimental to desired circuit behaviour. The steady state response of the circuit, which is determined by the external source, is reached only after a transient time interval. The above describe the time domain solutions.

Likewise, the current and voltage responses in circuits containing inductors and capacitors (besides resistors) are frequency-dependent because the impedances of the former elements vary with frequency. Such circuits can be therefore used as "frequency-selective circuits" or filters - that is, they respond to signals of different frequencies differently and can therefore be designed to "pass" certain frequencies to a load or other parts of the circuit and to attenuate others. The study of the frequency response of a network involves the consideration of the magnitude and phase of the steady state voltages and currents as the frequency varies. An integral part of such a study involves the concept of resonance - a condition in which a circuit response can be maximized by adjusting either the circuit elements or the frequency of operation.

This lab examines the behaviour of second-order circuits. You should complete the pre-lab exercises before coming to the lab.

2 Pre-Lab Exercises

1. Assume that the network in Figure 1 is initially at rest (zero state response)

   (a) Determine the transient response of v_{OUT} for $t \geq 0$ given that v_{IN} steps from 0 to v_T at $t = 0$. Note that, the response take the form of,

   $$v_{\text{OUT}} = v_{cp} + K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

   (b) Put $L = 47$ mH, $C = 0.1$ μF, $R_2 = 220$ Ω and $v_T = 10$ V, use Octave Online to graph the transient response of v_{OUT} for $0 \leq t \leq 10$ ms. On a separate graph, repeat the process for $R_2 = 2.2$ kΩ, and 4.7 kΩ.
   Octave Online is free and runs in a web browser at `https://octave-online.net`, so nothing needs to be installed. It uses the same commands as MATLAB.
   You may want to use the Octave command "linspace" to generate the x-axis values (i.e. time). For example, `t = linspace(0, .005)` or `linspace(0,0.005,101)`.
   In Octave, you type "help command" in the command window if you want to know what any command does. For example, type "help plot" and that will tell you what plot does, and how to use it.

   (c) For all the values of R_2, compute the frequency ω_d at which the transient response oscillates, and the rate α at which the transient response decays.

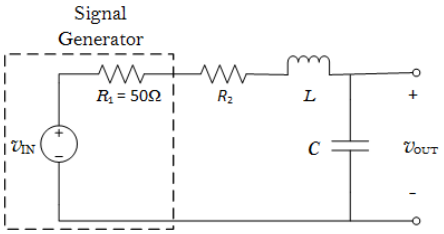


Figure 1: *RLC* Circuit for the Pre-Lab exercise 1

3 In-Lab Exercises

1. Construct the second-order network shown in Figure 1 using the 47 mH inductor, 0.1 μF capacitor and 220 Ω resistor.

   (a) Adjust the function generator to provide a 10 V peak-to-peak square wave at 200 Hz with a 5 V offset so that its open-circuit output voltage steps between 0 V and 10 V. Note that, this signal enables you to simulate the effect of repeatedly energizing the RLC circuit and the waveform in the oscilloscope may be considered similar to the step response of this circuit. Measure the voltage over the capacitor and take note of the transient response of the capacitor voltage $v_{\mathrm{OUT}}(t)$, and compare the response to that plotted during the pre-lab exercises.

   Also measure the voltage V_P (the first peak voltage of the transient response), the oscillation frequency ω_d, the rate of decay α. Measuring the α is the same as measuring the time constant τ over which the transient response has decayed by $1/e$ and then computing $\alpha = 1/\tau$.

   Replace the 220 Ω resistor with 2.2 kΩ, and 4.7 kΩ resistors and repeat the exercise.

4 Post-Lab Exercises

1. With reference to your In-Lab Exercise 1

   (a) Compare the response of the RLC circuits obtained from the experiment with the one found during the pre-lab exercise. Discuss the differences between the types of response for the three values of R_2. Compare the experimentally determined circuit characteristics with theoretical characteristics.